The Pharmacist's Cipher: Six Statistical Tests Supporting a Pharmaceutical Reading of the Voynich Manuscript (MS 408)

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AI Disclosure: Computational analysis and manuscript preparation were performed with the assistance of Claude (Anthropic), an AI language model.
All hypotheses, interpretive decisions, and epistemic judgments are the author's.

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Abstract

We present six reproducible statistical tests on the text of the Voynich Manuscript (MS 408, Yale Beinecke Library), based on the IVTFF interlinear transcription corpus. All tests are computed from the transcription alone, with no dependence on visual analysis of illustrations or assumed decipherment. The results are collectively consistent with a pharmaceutical interpretation of the manuscript's content: (1) morphological prefix frequencies are stable across two independent transcribers (all 11 tested prefixes within 4% divergence), establishing that the observed structure is a property of the manuscript, not of any single transcriber; (2) a binary morphological gap exists between two prefix families (*s-* nominal vs. *sh-* processual), with the token *sedy* attested exactly 0 times against *shedy* at 424, consistent with a systematic grammatical distinction between the two prefix families; (3) a 26:0 asymmetric lexical reuse pattern between the Pharmaceutical and Stars sections is compatible with functional dependency; (4) word-order analysis reveals a statistically significant tendency for completion markers to appear in the final 20% of lines ($Z = 26.6$); (5) six manuscript sections exhibit systematically distinct morphological profiles consistent with specialized functional registers; (6) the proportion of an infixed morpheme (*ee*) across solvent-prefix families replicates the pharmacologically expected volume hierarchy (water $>$ spirit $\approx$ oil $>$ vinegar $>$ medicine; Spearman $\rho = 0.900$, $p = 0.042$ one-tailed). These findings do not constitute a decipherment but establish that the Voynich text contains internally consistent, non-random structure that is consistent with a medieval pharmaceutical register.

Keywords: Voynich Manuscript, MS 408, computational linguistics, pharmaceutical codex, morphological analysis, medieval pharmacopoeia

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1 Introduction

The Voynich Manuscript (MS 408, Yale Beinecke Rare Book and Manuscript Library) is a 15th-century codex written in an undeciphered script, with elaborate botanical, astronomical, and balneological illustrations (D'Imperio, 1978). Despite over a century of cryptanalytic effort, no proposed decipherment has achieved scholarly consensus.

Recent computational approaches have shifted focus from attempted decipherment to structural analysis of the text itself. The availability of machine-readable interlinear transcriptions—notably the IVTFF format containing multiple independent transcriptions aligned to the same text—enables reproducible statistical testing of hypotheses about the manuscript's content.

This paper presents six statistical tests that collectively support a *pharmaceutical* interpretation of the manuscript. We do not claim to have deciphered the text. Rather, we demonstrate that the text's statistical properties are consistent with a medieval pharmaceutical register organized around solvent-based preparation methods, and harder to reconcile with some competing accounts (such as purely random or decorative text generation).

1.1 Methodological Principles

All analyses in this paper follow three principles:

1. **Reproducibility from transcription alone.** Every result can be computed from the publicly available IVTFF corpus file. No result depends on visual analysis of illustrations, assumed phonetic values, or proposed glosses.

2. **Standalone token matching.** Tokens are delimited by the dot separator in the transcription. All counts use exact standalone matching (word-boundary), never substring matching.

3. **Separation of structure, function, and gloss.** We distinguish three levels: *structural facts* (measurable distributional properties), *functional roles* (hypothesized roles within the system), and *glosses* (proposed semantic translations). We report primarily structural facts derived from transcriptional distributions. Some analyses group tokens into model-dependent functional categories for testing purposes; these groupings are made explicit and should not be mistaken for deciphered glosses.

1.2 Corpus

The corpus is the IVTFF interlinear transcription (Zandbergen, 2022), which contains multiple aligned transcriptions of the same text by different scholars. We use primarily the Takahashi transcription (filter code `;H>`) and, for cross-validation, the Currier transcription (filter code `;C>`). Tokens are separated by dots (`.`), not spaces. The full Takahashi corpus contains 37,036 tokens across 5,206 lines.

The manuscript's sections, identified by folio range, are: Herbal A (f1–f57), Herbal B (f58–f66), Zodiac (f67–f73), Balneo (f75–f84), Cosmo (f85–f86), Pharma (f87–f102), and Stars (f103–f116).

2 Result 1: Cross-Transcriber Stability of Morphological Prefixes

2.1 Rationale

Any structural claim about the manuscript's text must be robust to transcriber disagreement. Two independent transcribers—Takahashi (1998, code `;H>`) and Currier (1976, code `;C>`)—provide the opportunity for a direct stability test.

2.2 Method

We extracted all lines where both transcribers provide a reading for the same folio and line number, yielding 2,530 common lines (17,954 tokens from Takahashi, 17,481 from Currier). For each of 11 morphological prefixes, we computed the density (occurrences per thousand tokens, ‰) in each transcriber's text, then calculated the ratio C/H.

A prefix was counted when a standalone token (dot-delimited) begins with that character sequence. For *s*-, tokens beginning with *sh* were excluded to avoid conflation. For *t*-, tokens beginning with *th* were excluded.

2.3 Results

Table 1: Cross-transcriber prefix density comparison (2,530 common lines).

Prefix	**H (‰)**	**C (‰)**	**Ratio C/H**	**Drift**
ch-	161.4	166.0	1.028	2.8%
sh-	99.4	96.3	0.969	3.1%
da-	72.9	72.9	1.000	0.0%
ok-	51.8	52.6	1.015	1.5%
qok-	86.6	87.4	1.009	0.9%
qot-	32.6	33.6	1.029	2.9%
ot-	48.1	49.8	1.037	3.7%
l-	27.9	28.0	1.002	0.2%
y-	47.2	47.0	0.997	0.3%
s-	38.2	37.6	0.984	1.6%
t-	28.4	28.3	0.997	0.3%

All 11 prefixes fall within 4% divergence; 9 of 11 fall within 3%. The maximum drift is 3.7% (*ot*-). The ch/sh ratio per section—a key parameter in the pharmaceutical model—is stable: Herbal A shows H=2.17, C=2.26 (4.1% drift); Balneo shows H=1.07, C=1.11 (4.0% drift).

2.4 Interpretation

The morphological prefix structure of the text is a property of the manuscript itself, not an artifact of any single transcription methodology. This establishes that all subsequent analyses, which rely on prefix distributions, are robust to transcriber variation within the tested margin.

3 Result 2: Binary Morphological Gap Between *s-* and *sh-*

3.1 Rationale

If the text encodes a pharmaceutical system with distinct "cold process" and "ingredient/substance" categories, we would expect a morphological distinction between these two functions. Two prefix families in the transcription—*s-* (initial *s* not followed by *h*) and *sh-*—are candidates for such a distinction.

3.2 Method

We counted all standalone tokens beginning with *s-* (excluding *sh-*) and all tokens beginning with *sh-*, then classified each by its suffix: *processual* suffixes (*-edy*, *-eedy*, *-ey*, *-eey*) and *nominal* suffixes (*-aiin*, *-ain*, *-ar*, *-al*).

3.3 Results

Table 2: Paradigm gap: processual vs. nominal suffix distribution.

	s- tokens	**sh- tokens**	**Ratio sh/s**
Total tokens	1,250	3,175	2.54
Exact processual tokens			
sedy	**0**	—	—
shedy	—	**424**	∞
seo / sheo	0 /	47	∞
Comprehensive suffix categories			
Processual (-edy/-eedy/-ey/-eey)	93 (7.4%)	1,070 (33.7%)	4.6×
Nominal (-aiin/-ain/-ar/-al)	456 (36.5%)	333 (10.5%)	0.29×

The token *sedy* is attested exactly 0 times in the entire corpus (37,036 tokens). Its counterpart *shedy* occurs 424 times. This is not a statistical tendency but a categorical, binary gap: *s-* never takes the core processual suffix *-edy*; *sh-* takes it productively.

The gap extends to all processual suffixes: only 7.4% of *s-* tokens carry processual endings, compared to 33.7% of *sh-* tokens. Conversely, 36.5% of *s-* tokens carry nominal suffixes, compared to 10.5% for *sh-*—a 3.5× inversion. The 93 *s-* tokens with processual suffixes are predominantly compound forms (e.g., *solchedy*=8, *schedy*=7, *solkeedy*=5), where the processual suffix attaches to an intervening element, not directly to *s-*.

3.4 Interpretation

This binary gap is consistent with a system in which *s-* marks a nominal category (a substance or quality) while *sh-* marks a processual category (an operation or transformation). The categorical nature of the gap (*sedy*=0, not merely rare) argues against random variation and supports a grammatical distinction encoded in the script.

4 Result 3: Asymmetric Lexical Reuse Between Pharma and Stars Sections

4.1 Rationale

If the manuscript's sections serve different functional roles (e.g., base recipes vs. standardized preparations), we might expect asymmetric lexical reuse: tokens characteristic of one section appearing in another, but not the reverse.

4.2 Method

For each folio in the Pharma (f87–f102) and Stars (f103–f116) sections, we extracted the first token of the first line (the "header token"). We then searched the *other* section for exact standalone occurrences of each header token. This provides a mechanical, non-interpretive measure of cross-referencing: a section "cites" another if its header tokens appear as standalone words in that other section.

We tested 32 Pharma header tokens against 10,668 Stars tokens, and 23 Stars header tokens against 3,811 Pharma tokens.

4.3 Results

Table 3: Asymmetric cross-reference between Pharma and Stars.

Direction	**Occurrences**	**Distinct types**	**Denominator**
Pharma headers → found in Stars	26	7 of 32	10,668 tokens
Stars headers → found in Pharma	0	0 of 23	3,811 tokens

The asymmetry is total: 26 vs. 0. Seven of 32 Pharma folio headers appear as standalone tokens in Stars (e.g., *tchedy* from f94r appears 10 times in Stars; *pcheor* from f90v1 appears 4 times). Zero of 23 Stars folio headers appear in Pharma. The asymmetry persists despite the Stars section having 2.8× more tokens than Pharma, which would bias *against* the observed direction if cross-referencing were random.

4.4 Interpretation

This pattern constitutes asymmetric lexical reuse: header tokens from Pharma folios appear as standalone words in Stars, but not vice versa. The asymmetry is mechanical (based on exact token matching at fixed positions) and does not depend on semantic interpretation. It is compatible with a functional dependency in which Stars draws upon material defined in Pharma, but the test alone does not rule out other explanations for the asymmetry (e.g., differences in vocabulary size or token frequency between the two sections). A randomized baseline comparing the observed asymmetry against chance expectation would strengthen this result.

5 Result 4: Positional Constraint on Completion Markers

5.1 Rationale

Procedural texts in any language tend to place completion or closure markers at or near the end of instructions. If the Voynich text encodes procedural pharmaceutical content, tokens hypothesized as completion markers should cluster in line-final positions.

5.2 Method

We classified all tokens in procedural sections (Herbal A, Herbal B, Pharma, Balneo; 2,780 lines with $\geq$2 tokens, totaling 21,101 tokens) into four categories based on prefix: OPER (operational prefixes: *ch-*, *sh-*, *ot-*, *y-*), VEIC (vehicle/solvent prefixes: *qok-*, *qot-*, *ok-*, *sol*), MAT (material prefixes: *da-*, *k+*quadripole, gallows ornate), and COMPL (completion markers: *dam*, *sam*, *dy*, tokens ending in *-am*).

For each token, we computed its normalized position within its line (0.0=first token, 1.0=last token).

5.3 Results

Table 4: Mean normalized position by functional category.

Category	N tokens	Mean position
VEIC	3,644	0.478
OPER	6,943	0.484
MAT	2,537	0.582
COMPL	487	0.819

The strongest signal is the line-final clustering of COMPL tokens: 68.2% of completion markers appear in the final 20% of their line, against 20% expected by chance ($Z = 26.6$, $p < 10^{-10}$). The ordering VEIC$\approx$OPER $<$ MAT $<$ COMPL is consistent across sections.

Five of six pairwise directional tests are significant: operations tend to precede material (60.0% vs. 40.0%, $Z = 15.9$), and material tends to precede completion (78.2% vs. 21.8%, $Z = 11.4$). The one non-significant pair is OPER vs. VEIC (49.4% vs. 50.6%, $Z = -1.1$), consistent with these categories being positionally interchangeable.

5.4 Interpretation

The text exhibits a consistent word-order tendency: operations and vehicles appear before materials, and materials appear before completion markers. This pattern is compatible with verb-first procedural syntax (as found in medieval Arabic pharmaceutical texts, where the imperative opens the instruction and the completion formula closes it), but the positional data alone do not prove any specific linguistic affiliation.

Caveat on category assignment. The four functional categories (OPER, VEIC, MAT, COMPL) are model-dependent: they are defined by prefix membership, not by independently established grammatical criteria. The strongest positional signal—the line-final

clustering of COMPL ($Z = 26.6$)—is robust to reasonable variations in category boundaries (e.g., removing *dy* from COMPL still yields $Z > 15$ for the remaining tokens). Nevertheless, the result is conditional on the validity of these category assignments, and alternative groupings could in principle yield different patterns.

6 Result 5: Section-Specific Morphological Profiles

6.1 Rationale

If the manuscript's sections serve different functional purposes, their morphological profiles should differ systematically. A purely random or homogeneous text would show uniform prefix distributions across sections.

6.2 Method

For each of the six major sections, we computed the density (‰) of key morphological prefixes. To identify section-specific enrichment, we calculated the ratio of each section's density to the corpus-wide baseline.

6.3 Results

Table 5: Morphological prefix densities by section (‰). Bold values indicate ≥1.5× enrichment or ≤0.5× depletion relative to corpus baseline.

Prefix	**Herb. A**	**Herb. B**	**Zodiac**	**Balneo**	**Pharma**	**Stars**	**Corpus**
ch-	189.4	149.2	154.0	119.2	168.5	162.1	157.9
sh-	87.1	110.3	63.0	111.5	80.6	74.5	85.7
ch/sh	2.17	1.35	2.45	1.07	2.09	2.17	1.84
qok-	40.8	86.8	**13.5**	**157.0**	67.2	102.5	83.2
qot-	29.6	23.5	**4.0**	38.5	**13.9**	38.7	30.4
da-	91.7	83.5	62.6	50.1	76.4	32.4	60.9
ot-	42.4	51.1	**155.0**	45.3	40.9	74.2	63.7
l-	**9.5**	17.0	**4.0**	56.2	**9.2**	**73.2**	37.1
y-	65.2	70.6	72.9	**21.3**	50.4	33.0	48.3
N tokens	*9,595*	*1,233*	*3,033*	*6,886*	*3,811*	*10,668*	*37,036*

Key findings:

- **Balneo**: Near-parity of ch/sh (1.07) and maximal *qok-* enrichment (1.89× baseline). This is the only section where the two process-marker families are nearly balanced, and water-based prefixes are at their maximum. These distributional properties are compatible with a bathing or hydrotherapy context, but also with any register in which the relevant prefix families converge.
- **Zodiac**: Maximal *ot-* enrichment (2.43×) and near-zero *qok-*/*qot-* (depleted to 0.16× and 0.13×). The prefix profile is descriptive rather than procedural in character. This is compatible with astrological-medical content (melothesia), but the distributional data alone do not identify the subject matter.

- **Stars**: Maximal *l*- enrichment (1.97× baseline), the highest in the corpus. If *l*- marks a specific solvent (as suggested by its distributional profile elsewhere in the corpus), this enrichment would be compatible with a formulary of concentrated preparations. However, the functional identification of *l*- is model-dependent.
- **Pharma**: Highest ch- density (168.5‰) and relatively low *ot*- (40.9‰). Active process markers dominate over passive constructions. This is compatible with procedural instructional content, though the interpretation relies on the proposed functional categories.

6.4 Interpretation

The six sections exhibit systematically distinct morphological profiles. The variation is not random: specific prefixes are enriched or depleted in section-specific patterns. Each section's profile is compatible with a specific functional role within a pharmaceutical system—botanical description (Herbal), astrological properties (Zodiac), therapeutic application (Balneo), base preparations (Pharma), and standardized formulations (Stars)—but we note that these functional labels are interpretive readings of the distributional data, not conclusions entailed by the data alone.

7 Result 6: Morpheme Frequency Replicates Pharmacological Volume Hierarchy

7.1 Rationale

In a pharmaceutical text, the frequency of "abundant quantity" markers should correlate with the typical physical volume of each solvent. Water is used in liters (washes, decoctions), oil in spoonfuls (ointments), spirit in drops (tinctures), vinegar in small quantities (medicated vinegars), and pure medicine in minimal doses. If the morpheme *ee* functions as a quantity marker, its relative frequency across solvent families should replicate this hierarchy.

7.2 Method

For each solvent-prefix family, we counted the total number of standalone tokens beginning with that prefix and the number of those tokens containing the bigraph *ee*. We computed the percentage $\text{ee\%} = N_{\text{ee}}/N_{\text{total}} \times 100$. We then compared the observed ranking with the pharmacologically expected volume ranking using Spearman's rank correlation ($n = 5$, exact permutation test with all 120 permutations enumerated).

Five solvent families were defined: WATER (*ok*- and *qok*-), OIL (*t*- excluding *th*-, and *qot*-), SPIRIT (*l*-), VINEGAR (*s*- excluding *sh*-), and MEDICINE (*da*-).

Disclosure: The hypothesis that *ee* marks abundant quantity, and that its frequency should correlate with solvent volume, was formulated after preliminary observation of the data. This test is therefore exploratory rather than confirmatory, and the p-value should be interpreted accordingly.

7.3 Results

Table 6: Proportion of tokens containing *ee* by solvent-prefix family.

Solvent family	Total	with *ee*	ee%	Expected rank	Observed rank
WATER (ok+qok)	5,505	1,278	23.2%	1	1
SPIRIT (l)	1,374	210	15.3%	3	2
OIL (t+qot)	2,098	279	13.3%	2	3
VINEGAR (s)	1,250	56	4.5%	4	4
MEDICINE (da)	2,256	10	0.4%	5	5

Spearman $\rho = 0.900$ ($\Sigma d^2 = 2$, $n = 5$). Exact permutation test: $p = 0.042$ one-tailed, $p = 0.050$ two-tailed.

The only rank inversion is between OIL and SPIRIT (13.3% vs. 15.3%), which are pharmacologically adjacent in expected volume. At the individual prefix level, the dosage-specific forms show a finer-grained pattern: *qok-* (25.5%) > *ok-* (20.3%) > *qot-* (15.6%) ≈ *l-* (15.3%) > *t-* (10.7%) > *s-* (4.5%) > *da-* (0.4%).

7.4 Interpretation

The observed ranking is compatible with the pharmaceutical volume hypothesis: water-related tokens carry the *ee* bigraph most frequently, medicine-related tokens almost never. The correspondence appears difficult to explain under a purely content-neutral account, but the exploratory status of the test requires caution.

The single rank swap (oil and spirit) is pharmacologically plausible: both solvents are used in comparable moderate quantities, and their near-identical ee% (13.3% vs. 15.3%) reflects this similarity.

Important caveats. This test is exploratory: the hypothesis that *ee* marks abundant quantity, and that its frequency should correlate with solvent volume, was formulated after observing the data. The two-tailed $p = 0.050$ is at the conventional significance boundary with $n = 5$. The solvent families are not perfectly homogeneous (e.g., WATER pools *ok-* and *qok-*, which may have partially distinct functions). Independent replication on a held-out corpus partition or comparison with known pharmaceutical texts would be needed to elevate this result from exploratory to confirmatory.

8 Discussion

8.1 Summary of Evidence

The six results converge on a single picture: the Voynich Manuscript's text contains internally consistent, non-random structure that is compatible with a medieval pharmaceutical register. Specifically:

1. The structure is *real*: it survives cross-transcriber validation (Result 1).

2. The structure is *grammatical*: it exhibits categorical morphological gaps (Result 2).

3. The structure is *organized*: different sections exhibit asymmetric lexical reuse (Result 3).

4. The structure is *procedural*: completion markers cluster at line endings (Result 4).

5. The structure is *functionally differentiated*: sections have distinct morphological signatures (Result 5).

6. The structure is *domain-compatible*: a morpheme's distribution replicates a pharmacological constraint (Result 6).

8.2 Limitations and Alternative Hypotheses

Several important limitations must be acknowledged.

Internal coherence is not proof. A sufficiently expressive morphological model can accommodate many datasets. Our results demonstrate that the pharmaceutical hypothesis is *consistent* with the data, not that it is the *only* hypothesis consistent with the data.

Semi-independence of tests. Results 2, 4, 5, and 6 share underlying assumptions about prefix classification. Although each test measures a different property (categorical gap, positional distribution, sectional variation, morpheme frequency), they are not fully independent. Result 1 (cross-transcriber) and Result 3 (cross-reference asymmetry) are the most independent, as they rely on mechanical matching rather than interpretive categories.

Exploratory analysis. Result 6 was developed after observing patterns in the data (post-hoc). While we report exact p-values from permutation tests, these should be interpreted in the context of exploratory analysis. The remaining five results test pre-specified structural properties and do not carry this caveat.

The null hypothesis is imprecise. We have not formally specified what a "non-pharmaceutical" text would look like. Our comparisons are with pharmacological expectations, not with explicit null models. Future work should compare the observed distributions with known texts (pharmaceutical and non-pharmaceutical) in medieval Arabic and Latin traditions.

The astronomy alternative. Several distributional properties we attribute to pharmaceutical content (e.g., section-specific profiles, processual morphology) could also be compatible with an astronomical or astrological register. The Zodiac section's enrichment in passive-reflexive constructions (Result 5) is consistent with *both* pharmaceutical astrology (melothesia) and general astrology. Result 6 (volume ranking) is the most discriminating, as it has no natural astronomical interpretation.

8.3 Relation to Prior Work

Our approach builds on quantitative studies of the Voynich text by Currier (1976), D'Imperio (1978), Stolfi (2005), and Amancio et al. (2013), which established that the text has non-random statistical properties including Zipf-law compliance, low entropy, and section-specific "languages." Our contribution is to show that these properties are specifically consistent with a pharmaceutical register, rather than with natural language in general.

The pharmaceutical hypothesis itself draws on the work of Bax (2014), who proposed partial phonetic readings of plant names, and on the broader tradition of interpreting

the manuscript's botanical illustrations as depicting medicinal plants (Tucker & Talbert, 2013; Sherwood & Schwartz, 2014). Our statistical approach complements these visual and phonetic methods by providing distributional evidence that does not depend on assumed sound values or plant identifications.

8.4 Predictions and Future Work

The pharmaceutical framework generates testable predictions for future work:

1. **Visual validation.** If the manuscript's illustrations encode pharmacological information through color (as preliminary visual analysis suggests), the color distributions should correlate with the textual morphological profiles reported here.

2. **Cross-linguistic comparison.** The section-specific morphological profiles (Result 5) should match more closely with medieval Arabic/Persian pharmaceutical texts (*aqrābādhīn* genre) than with astronomical or literary texts.

3. **Additional paradigm gaps.** If the *s-*/*sh-* gap reflects a systematic cold/nominal distinction (Result 2), analogous gaps should exist for other claimed morphological pairs in the system.

4. **Volume ranking stability.** The ee% volume ranking (Result 6) should remain stable when computed separately for each section, with possible attenuation in sections where solvent volumes are contextually implicit (e.g., Balneo, where water is the ambient medium).

A v2 of this preprint, including high-resolution illustrations from the Yale IIIF archive and formal comparison with medieval pharmaceutical corpora, is in preparation.

9 Conclusion

Six reproducible statistical tests on the Voynich Manuscript's text are consistent with a pharmaceutical interpretation of its content. The text exhibits real (cross-transcriber stable), grammatical (paradigm gap), organized (asymmetric lexical reuse), procedural (positional constraint), differentiated (section-specific profiles), and domain-compatible (volume hierarchy) structure. These properties are difficult to reconcile with a purely meaningless or decorative account and are specifically compatible with a medieval pharmaceutical register organized around solvent-based preparation methods.

We emphasize that compatibility is not proof. The decipherment of the Voynich Manuscript remains an open problem. What we offer is a quantitative framework that narrows the hypothesis space and generates falsifiable predictions for future work.

Data Availability

The IVTFF interlinear transcription corpus is publicly available at `https://www.voynich.nu/`. A companion repository containing all analysis scripts (Python), regular expressions used for prefix extraction, the exact token lists for each functional category, and the folio-to-section mapping is available at `https://github.com/alessandroplaca-uro/voynich-pharmacists-cipher`.

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